Smrithi Lokesh Seramalanna

smrithi@seas.upenn.edu | https://www.linkedin.com/in/smrithi-lokesh-0894b7219 | https://smrithilo.github.io/smrithi.github.io/

Professional Summary

ML/AI Engineer with a robotics systems foundation, pursuing an MSE in Robotics at the University of Pennsylvania. Research focus spans robot learning, visuomotor imitation learning, and autonomous perception — with hands-on experience training diffusion policies on real hardware, curating multimodal datasets, and building full autonomy stacks from sim through deployment. Published at IROS 2026. Seeking roles at the intersection of machine learning and physical robotics systems.

Education

University of Pennsylvania
MSE Robotics. *Expected: December 2027*
Relevant Coursework: CIS 5200 Machine Learning, CIS 5800 Machine Perception, MEAM 6200 Advanced Robotics

Birla Institute of Technology and Science, Pilani
B.E. (Hons.) Mechanical Engineering, Minor in Robotics and Automation. GPA: 9.14/10.00. *July 2025*

Experience

Research Assistant May 2026 – Present
Janus Intelligent Robot Lab, GRASP, University of Pennsylvania *Philadelphia, PA*

- Developing visuomotor behavioral-cloning pipelines for bimanual manipulation on UR5 robots, using wrist-camera observations and diffusion policies to automate multi-step chemistry laboratory tasks.
- Curating multimodal demonstration datasets by filtering low-information plateaus, incorporating action history and DAgger corrections, and investigating joint training across multiple task policies to improve execution success; research ongoing toward an ICRA 2027 submission.
- Extending the system to long-horizon manipulation tasks, debugging sim-to-real and hardware failures, and scaling data collection and policy training across tasks.

Robotics Engineer Aug 2025 – Dec 2025
Open Science Stack *Bengaluru, India*

- Built and deployed foundational autonomy infrastructure for UR and xArm robots in a pharma-lab-equivalent setting, enabling subsequent automation workflows at an early-stage robotics startup.
- Migrated the robotics stack from ROS to ROS 2, switched simulation to Isaac Sim, and implemented GPU-accelerated motion planning using cuMotion integrated with MoveIt2.

Research Intern Jul 2024 – Jan 2025
Carnegie Mellon University *Pittsburgh, USA*

- Developed a hierarchical planner for the Cuboid Multi-Agent Collective Construction (Cuboid-MACC) problem using graph-search approaches to determine sequences of abstract block placements and removals.
- Verified executability by computing reachable access poses and encoding action dependencies as an acyclic Action Dependency Graph.
- Extended Conflict-Based Search (CBS) to resolve multi-robot path conflicts in dynamic environments, accounting for rotational asymmetry of cuboid blocks and carried-block collision constraints; validated on 200 randomly generated structures.

Research Intern May 2024 – Jul 2024
Indian Institute of Science x Cornell University *Bangalore, India*

- Researched soft robot stiffness in the Ruina Lab (Prof. Andy Ruina), designing and fabricating fiber-reinforced, pneumatically actuated soft bodies.
- Engineered an experimental setup to analyze how transmitted forces vary under controlled pressure, volume, and control-cable tensioning.

Research Intern May 2023 – Jul 2023
Defence Research and Development Organisation *Hyderabad, India*

- Conducted experimental characterization of carbon-epoxy composites for rocket motor casings in the Advanced Systems Laboratory under Dr. Lokesh Srivastava, Scientist F.
- Performed reliability prediction of composite samples using linear curve-fitting and Weibull analysis.

Exchange Student Jan 2025 – Jun 2025
Mälardalens University *Västerås, Sweden*

- Recipient of the European Union Erasmus+ ICM Scholarship; enrolled in advanced robotics coursework specializing in planning, perception, and autonomous systems.

Projects

Autonomous VIO-Based Quadcopter Navigation Spring 2026
MEAM 6200 Advanced Robotics

- Integrated A* planning on a voxelized map, minimum-snap trajectory generation, SE(3) geometric control, and an Error-State Kalman Filter (ESKF) for IMU/stereo-visual fusion into a full closed-loop autonomy stack.
- Achieved safe navigation across six obstacle environments (maze, slalom, stairwell, switchback, window, over/under) using only on-board VIO state estimates; co-designed trajectory aggressiveness with estimator bandwidth to prevent visual feature dropout.

Two-View Stereo 3D Reconstruction Spring 2026
CIS 5800 Machine Perception

- Implemented a complete stereo pipeline: rectification, disparity map computation (SSD, SAD, ZNCC with L-R consistency), depth estimation, and back-projected point-cloud generation in world frame.
- Aggregated multi-pair reconstructions to produce dense 3D point clouds of real scenes.

Aircraft Design and Analysis September 2022 – April 2024

- Designed and built a micro-class fixed-wing aircraft demonstrating Urban Air Mobility (UAM) missions for AIAA Design, Build and Fly 2024; ranked 26th of 107 internationally.

Awards & Honors

- Erasmus+ ICM Scholarship, European Union
- Overall Excellence Award, International Rover Challenge 2023
- 26th of 107 Internationally, AIAA Design, Build and Fly 2024
- Partial Travel Grant, International Programs and Collaborations Division, BITS Pilani

Skills

ML and AI: PyTorch, Diffusion Policies, Behavioral Cloning, OpenCV, Machine Learning, Computer Vision

Programming Languages: Python, C++, MATLAB/Simulink, Java

Robotics and Simulation: ROS 2, MoveIt2, cuMotion, NVIDIA Isaac Sim, Gazebo, RViz

Hardware: UR3e, UR5, Franka Emika Panda, xArm, Robotiq Grippers, Arduino, Raspberry Pi, 3D Printing

CAD and Design: SolidWorks, Fusion 360, XFLR5

Publications

- Hierarchical Planning for the Cuboid Multi-Agent Collective Construction (Cuboid-MACC) Problem — Junil Min, Shambhavi Singh, Geordan Gutow, Smrithi Lokesh Seramalanna, Bhaskar Vundurthy, Jinwoo Choi, Howie Choset. Accepted to IROS 2026 (IEEE/RSJ International Conference on Intelligent Robots and Systems). Jun 2026.
- Kinematic models for Cable-driven Continuum Robots with multiple segments and varying cable offsets — Ashish Bhalkikar, Smrithi Lokesh, Ashwin K P. Mechanics and Machine Theory, doi.org/10.1016/j.mechmachtheory.2024.105701. Sept 2024.

Teaching And Volunteering

- Lab Staff, Additive Manufacturing Lab (AddLab), University of Pennsylvania (May 2026 – Present): Support research labs and student engineering projects through additive manufacturing, rapid prototyping, and fabrication; operate and maintain FDM, SLA, and resin-based 3D printing systems.
- Teaching Assistant – Artificial Intelligence, Engineering Summer Academy, University of Pennsylvania (July 2026): Supported Prof. Robert Ghrist in a summer program introducing high-school students to foundational AI and algorithmic thinking.
- Scratch Instructor, Fife-Penn STEM and CS Academy, St. Francis Xavier School (January 2026 – Present): Teach foundational programming concepts to K-8 students.
- Undergraduate Teaching Assistant – Mechanics of Solids, BITS Pilani (Aug 2023 – Dec 2023): Addressed student queries, evaluated quizzes, and managed academic records for 180 sophomore students under Prof. Kiran Dinkar Mali.
- Instructor – Aircraft Design Course, Centre of Technical Education, BITS Pilani (Nov 2022 – Jan 2023): Conducted a short course on UAV performance and design covering airfoil and wing analysis, design parameter optimization, and CAD.